

Game idea: A level based puzzle game about making drinks, where the player must make the pictured drink using different ingredients and machines. Every drink starts as a plain glass of water. The player must discover and remember the rules (what happens when they use the ingredients and machines) and use their knowledge to do everything in the right order to find the correct solution. The way things interact is unknown to the player at the beginning but every combination leads to the same outcome every single time (ex. mixing yellow and blue makes green every time they are mixed).

Pillars:

- Problem solving
- Discovery
- Memory

Core loop:

Experiment -> Discover a rule -> Use knowledge -> Take action

Ingredients:

- Red syrup
- Yellow syrup
- Blue syrup

Machines and actions:

- Heating - lightens colour, makes bubbles bigger
- Cooling - darkens colour, makes bubbles smaller
- Carbonating - adds bubbles

End result of using multiple machines:

1st/2nd	Heating	Cooling	Carbonating
Heating	drink evaporates	back to medium colour	light with medium bubbles
Cooling	back to medium colour	drink freezes	dark with medium bubbles
Carbonating	light with big bubbles	dark with small bubbles	-

Heating and cooling only change colour by one step and carbonating adds bubbles only once. If heating an already heated drink or cooling an already cooled drink the player must restart, as the drink is ruined. Carbonation can only be added once, the machine can't be used twice on the same drink.

Impossible results:

- dark with big bubbles
- light with small bubbles

Scaling difficulty:

Machines and ingredients could be added incrementally to add difficulty by making more possible outcomes and combinations. This adds to the number of steps that must be done in the correct order.

Colour mixing chart:

Mixed colours			
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			-

Adding one colour twice has no effect. Example: red+red=red, red+red+yellow=orange. It does not matter how many times a colour is added. Mixing all three colours makes brown which means the player has to restart the level.

Colour lightness and darkness

Normal	Heating	Cooling

Example level (watch video):

Goal: dark green drink with medium bubbles

Correct solutions:

- 1) yellow -> blue -> cooling -> carbonating
- 2) blue -> yellow -> cooling -> carbonating

Possible mistakes:

- 1) carbonating before cooling
- 2) carbonating before adding syrups